Day 2: Pehli Gemini API Call

Gen-AI + Agentic AI with JavaScript (Gemini Edition) | Phase 0: Setup

1. Concept

API, SDK aur API Key

Term	English	Hinglish
API	A way for your code to talk to a remote service	Dusre server se baat karne ka rasta, jaise restaurant ka waiter
SDK	A ready-made library that wraps the API	Waiter ke saath menu card, taaki raw HTTP na likhna pade
API Key	A secret password that identifies you	Tumhari pehchaan aur quota ka hisaab, isko leak mat karna

Analogy: API = waiter, SDK = menu card, API key = membership card. Card kho gaya toh koi aur tumhare naam pe order karega.

Google AI Studio tour (10 min)

aistudio.google.com me ye 4 cheezein dekho:

1. **Get API key:** yahan se key banti hai
2. **Playground / Chat:** bina code ke prompts try karo (model, temperature, system instructions badal ke dekho)
3. **Get code:** Playground me chalaye prompt ka JS code mil jata hai
4. **Dashboard > Usage / Rate limits:** apni quota aur limits yahin dikhti hain. Free tier limits model ke hisaab se alag hoti hain, isliye yahin check karo.

Gemini SDK

- Node.js v18+ me `npm install @google/genai` se SDK install hota hai.
- Agar API key `GEMINI_API_KEY` environment variable me hai, toh client usse apne aap utha leta hai.
- Google ki docs ab `generateContent` ko "legacy" bol rahi hain aur naye projects ke liye Interactions API recommend karti hain. Lekin generateContent fully supported hai aur LangChain.js, Vercel AI SDK jaise tools abhi isi pe bane hain. Isliye pehle isse seekhenge, baad me Interactions API dekhenge. Concepts dono me same hain.
- Docs ke examples me `gemini-3.5-flash` use ho raha hai. Model names badalte rehte hain aur free tier me sab available nahi hote, isliye AI Studio me jo Flash model tumhari key pe chale wahi use karo.

Request aur Response structure

```
Request:  model + contents (tumhara prompt) + config (optional)
Response: text + usageMetadata (kitne tokens lage) + finishReason
```

2. Workflow

```
1. AI Studio me API key banao
2. .env me key rakho  (GEMINI_API_KEY=...)
3. .gitignore me .env daalo  (push karne se PEHLE)
4. npm install @google/genai dotenv
5. Client banao  -> new GoogleGenAI({})
6. ai.models.generateContent({ model, contents })
7. response.text print karo
8. response.usageMetadata se tokens dekho
```

Safety checklist

- .env **.gitignore** me hai
- Key code me hardcode nahi hai
- Repo me **.env.example** hai (sirf GEMINI_API_KEY=your_key_here)
- **Key push ho gayi?** Turant AI Studio se key delete karke nayi banao. Git history se hata dena kaafi nahi hota.

3. Code

Setup

```
cd phase-1-foundations && mkdir day-02 && cd day-02
npm init -y            # "type": "module" add karna mat bhoolna
npm install @google/genai dotenv
```

.env (kabhi commit mat karna):

```
GEMINI_API_KEY=paste_your_key_here
```

.env.example (ye commit hoga):

```
GEMINI_API_KEY=your_key_here
```

config.js: model ek hi jagah

```
// Model name yahin rakho, taaki badalna ho toh ek jagah se badle
export const MODEL = "gemini-3.5-flash"; // AI Studio me jo available ho wo daalo
```

first-call.js: pehli call

```
import "dotenv/config";
import { GoogleGenAI } from "@google/genai";
import { MODEL } from "./config.js";

// Fail fast: key na ho toh pehle hi bata do
if (!process.env.GEMINI_API_KEY) {
  console.error("GEMINI_API_KEY missing hai. .env file check karo.");
  process.exit(1);
}

// Key GEMINI_API_KEY env variable se apne aap utha leta hai
const ai = new GoogleGenAI({});

async function main() {
  const response = await ai.models.generateContent({
    model: MODEL,
    contents: "Explain what an API is in 2 lines, Hinglish me.",
  });

  console.log("Answer:", response.text);
  console.log("Tokens:", response.usageMetadata); // Day 3 me detail
}

main().catch((err) => {
  console.error("Call fail hui:", err.message);
  process.exit(1);
});
```

Run: `node first-call.js`

sentiment.js: Day 1 ki problem ab Gemini se solve

```
import "dotenv/config";
import { GoogleGenAI } from "@google/genai";
import { MODEL } from "./config.js";
```

```
const ai = new GoogleGenAI({});

const reviews = [
  "This phone is amazing, I love it",
  "The food was not good",
  "Yeh movie ekdum bakwaas thi",
  "Great, another delayed flight...",
];

for (const review of reviews) {
  const response = await ai.models.generateContent({
    model: MODEL,
    contents: `Classify the sentiment as positive, negative or neutral.
Reply with only one word.
Review: "${review}"`,
  });
  console.log(`"${review}" -> ${response.text.trim()}`);
}
```

Expected: ab "not good", Hinglish aur sarcasm teeno sahi pakde jayenge, bina ek bhi rule likhe. Loop me 4 calls ja rahi hain, aur free tier me RPM/RPD limits hoti hain, isliye bina soche loop me bahut saari calls mat chalana.

Common errors

Error	Matlab	Fix
`API key not valid`	Key galat ya expired	AI Studio se dobara copy karo
`429`	Rate limit ya quota khatam	Thoda ruko, limits check karo (Day 8 me retry seekhenge)
`404 model not found`	Model name galat ya available nahi	AI Studio me available model dekho, config.js badlo
`Cannot use import`	ES modules on nahi	package.json me "type": "module"

Homework

1. Code **khud type karo**.
2. sentiment.js me 2 aur tricky reviews daalo aur Day 1 ke rule-based.js se compare karo.
3. contents me alag prompts try karo aur dekho usageMetadata me tokens kaise badalte hain.
4. .env.example bana ke push karo, aur confirm karo ki .env GitHub pe nahi gaya.

4. Quick Revision (Quiz Answers)

Question	Answer
Analogy me SDK kis jaisa hai?	Menu card jo order dena aasan banata hai
API key kahan rakhna sahi hai?	.env file me, aur .env ko .gitignore me daalke
new GoogleGenAI({}) bina key kaise chalta hai?	SDK GEMINI_API_KEY environment variable apne aap padh leta hai
429 error ka matlab?	Rate limit ya quota cross ho gaya
Model name config.js me kyun?	Model names badalte rehte hain, ek jagah se change karna aasan hota hai

Next: Day 3: How an LLM works: tokens, context window, next-token prediction